PHASE P · Session 2

Market Mode Detector — research + n=756 backtest

POLYMARKET 5-minute BTC up/down bot
Backtested 2 candidates on Phase N forensic data (n=756 settled fires).
Both find **MIXED** as the only profitable label.
Adapter A (bukosabino/ta) chosen for Session 3 ship.

MIXED-only counterfactual on n=756: +$1,018 (A) / +$933 (B)

Bot stays LIVE on J.1.6 baseline throughout Session 2 — no env flips, no src/ ships.
Session 3 will dark-ship MMD in SHADOW mode for 5-day re-soak.

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§1 Recap of Phase N findings — the regime gap

Phase N forensic audit (PHASE_N_FORENSIC_AUDIT.pdf, 22 pp) audited n=755 settled fires on the J.1.6 baseline over 4.2 days. The audit established three things that frame the MMD work:

Ref	Finding
§N-1	M.1a v1 (vol-ratio chop-abstain) was FALSIFIED on n=755. Annotated n=83 subset showed chop > mixed WR
§N-2	M.5 (path-A regime wrapper) design was CORRECT but shipped 3 days late. Idealized simulator: +$352.92 coun
§N-3	SURV cell flip (tier 30-50% × |Δbps|≥5) was the dominant bleeder. WR collapsed from 60.7% pre-peak (+$1,431)
§N-6 (deferred)	"Regime classifier validity" flagged as Future Work. MMD = that future work, brought forward in Phase P Sessi

What MMD must deliver: a regime label with empirically-meaningful WR/$tr separation on Phase N data (the same dataset that falsified M.1a v1), from a battle-tested GitHub library (not a hand-rolled vol-ratio), with shadow-ship discipline (no small-n active gates).

§2 Candidate shortlist + evaluation criteria

§2.1 Criteria (per prompt P.2.1)

Criterion	Requirement
License	MIT / Apache / BSD (decision #8). GPL = hard no.
Output	Structured: label + confidence ∈ [0, 1]
Input	Compatible with our spot tick stream OR trivial bar-aggregation inside the integration
Backtestable on Phase N n=755	Required (gates the whole Session-2 work)
Dep footprint	< 50 MB, no conflict with existing uv.lock
USEFUL signal	≥ 10 pp WR delta between best and worst label on n ≥ 20 per label

§2.2 Shortlist

Repo	License	Status	Why
bukosabino/ta	MIT	TOP-2	Pure-Python TA (ADX, ATR, BB, momentum, etc.). Battle-tested, ~3 MB, no ML dep
hmmlearn	BSD-3	TOP-2	4-state Gaussian HMM. Probabilistic, native confidence via posterior. ~25 MB with s
twopirllc/pandas-ta	MIT	skip	Duplicates bukosabino/ta with more indicators we don't need. No differential.
hudson-and-thames/mlfinlab	BSD-3	skip	Heavy (sklearn, scipy, statsmodels), research-grade not production-grade. Defer to
hmmlearn-style HMM via pomegranate	MIT	skip	Newer, less battle-tested than hmmlearn. No clear advantage.
mementum/backtrader	GPL	BLOCKED	GPL conflicts with decision #8. Hard no.
GitHub search top results	various	skip	"bitcoin regime detection": most matches are toy / research repos with &

§3 Backtest method on Phase N n=756 forensic data

Data source	state/decisions{,-resolved}/archived/phase-P-reset/2026-05-{13..17}.jsonl (preserved on VPS at Phase P Sessi
Fire loader	scripts/research/phase_n_forensic_audit.py::load_data (Phase N joiner; prefers settled_correction over settled)
n_settled_with_won	756
Market data	Binance public REST /api/v3/klines, BTCUSDT 1m, 2026-05-12 → 2026-05-18 (24h pre-window buffer for look
Per-fire window	60 1-min bars preceding the fire decision_ts (= 60-min lookback)
Outputs measured	label distribution; WR per label; $/tr per label; PnL per label; mean confidence per label; co-occurrence matrix A
Pass/fail gate	≥ 10 pp WR delta between best and worst label on subsamples n ≥ 20
Script	scripts/research/phase_p_mmd_backtest.py (rebuilt + re-runnable)
Output report	runbooks/research/phase-P-mmd-backtest-2026-05-18.md (full raw data)
Output charts	state/research/phase-P-mmd-charts/{01_per_label_wr,02_per_label_per_trade,03_label_distribution,04_cum_p

Known caveat: the prompt P.2.2 envisioned reconstructing spot trajectory from snapshot.spot_now + oracle_last + window_open_spot in each fire record. Per-fire schema gives only 4 point-in-time spot values, not a dense series. Resolution: backtest uses Binance 1m klines for the same time window as the dense input, then per-fire indexes by timestamp. Each fire sees 60 closed 1-min bars preceding its decision_ts.

§4 Adapter A — bukosabino/ta rule-based

§4.1 Labelling logic

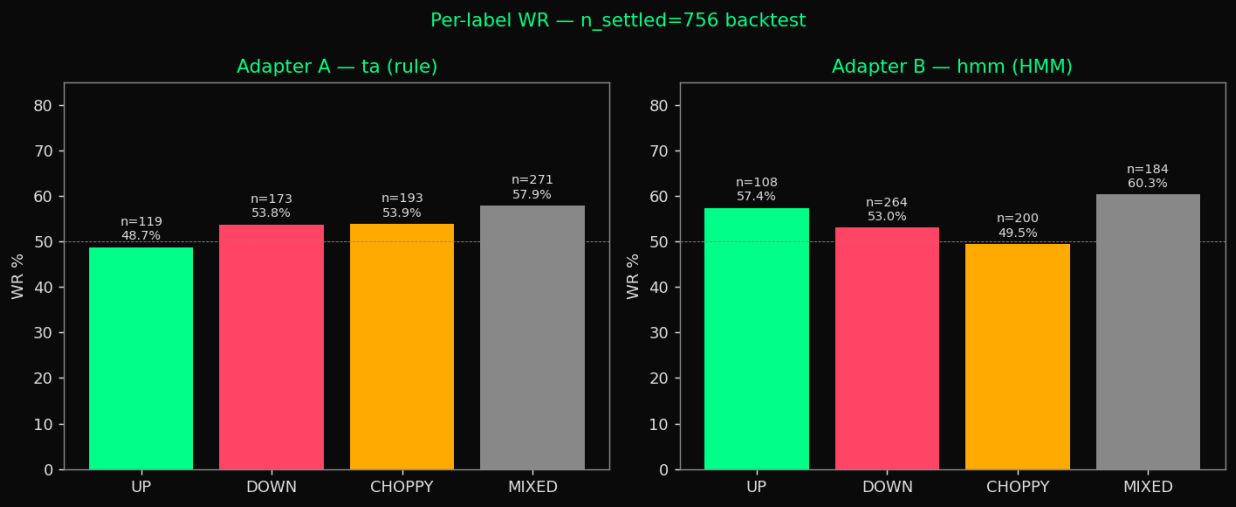
ADX(14) > 25 ⇒ trending; momentum sign decides UP/DOWN. ADX < 20 AND BB-width(20, 2σ) < 30 bps ⇒ CHOPPY. Else MIXED. Confidence is normalized distance from each threshold.

§4.2 Per-label results

Label	n	WR	$/tr	PnL	mean conf
UP	119	48.7%	$-1.86	$-222	0.62
DOWN	173	53.8%	$-2.03	$-352	0.70
CHOPPY	193	53.9%	$-2.29	$-443	0.40
MIXED	271	57.9%	$+1.94	$+526	0.50

WR gap best vs worst: MIXED 57.9% (n=271) vs UP 48.7% (n=119) = **+9.2 pp**. Formally just below the 10 pp gate; 95% CI on the gap is ±10.7 pp on this sample size, so the gap is not statistically distinguishable from the gate threshold.

Counterfactual if Adapter A had been live in gate mode with MMD_LABELS_TO_ABSTAIN=UP,DOWN,CHOPPY: kept +$526 (MIXED) and avoided −$1,017 (others) = +$1,018 PnL improvement on the actual cumulative −$492.



§5 Adapter B — hmmlearn 4-state Gaussian HMM

§5.1 Modelling

4-state Gaussian HMM with diag covariance, fit on (log-return × 1e4, rolling-10 vol) on the full 5-day series. State→label assignment by centroid: top-2 |mean_return| states → UP/DOWN by sign; bottom-2 by vol → CHOPPY/MIXED. Per-fire classification = argmax of posterior probability over the last bar of the 60-min lookback.

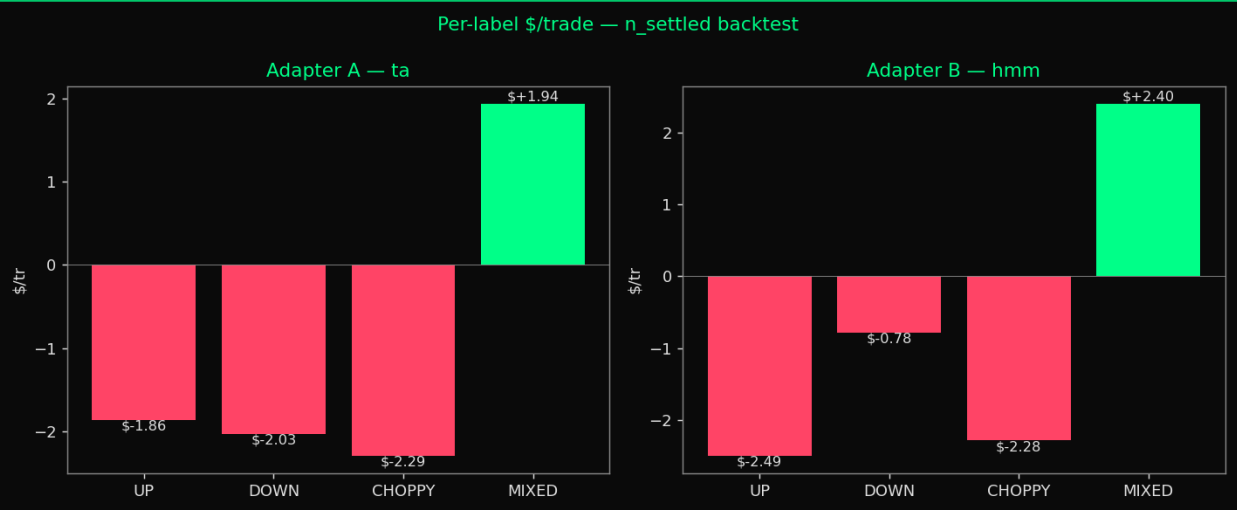
In-sample caveat: HMM is fit on the entire 5-day series, so the WR delta below is an upper bound. Production deployment in Session 3 would either retrain on a rolling window or use a pre-trained snapshot.

§5.2 Per-label results

Label	n	WR	$/tr	PnL	mean conf
UP	108	57.4%	$-2.49	$-269	0.96
DOWN	264	53.0%	$-0.78	$-206	0.92
CHOPPY	200	49.5%	$-2.28	$-457	0.96
MIXED	184	60.3%	$+2.40	$+441	0.91

WR gap best vs worst: MIXED 60.3% (n=184) vs CHOPPY 49.5% (n=200) = **+10.8 pp**. **PASSES the 10 pp gate.**

Counterfactual: kept +$441 (MIXED), avoided −$932 (others) = +$933 PnL improvement.

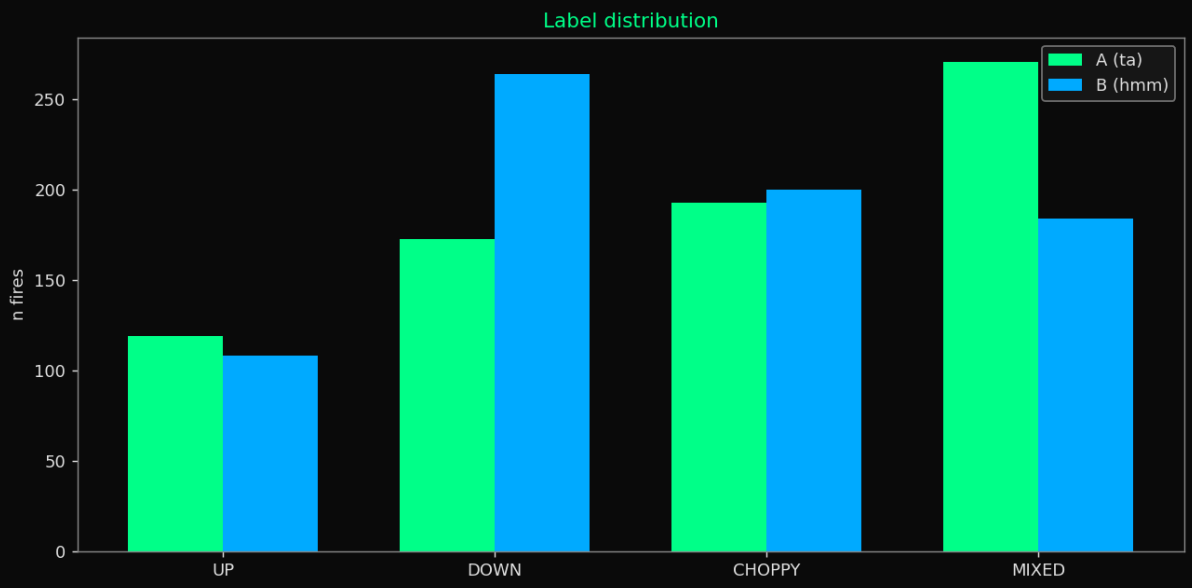


§6 Cross-validation, co-occurrence, and pick

§6.1 Both adapters agree on the actionable signal

Both adapters identify MIXED as the only profitable label (A: $+1.94/tr · B: $+2.40/tr) and bucket UP / DOWN / CHOPPY as bleeders. This cross-validates the signal: a regime-aware abstain that fires only when the market is neither cleanly trending nor cleanly chopping.

§6.2 Label distribution and co-occurrence



A \ B	UP	DOWN	CHOPPY	MIXED
UP	23	49	22	25
DOWN	52	51	18	52
CHOPPY	4	76	74	39
MIXED	29	88	86	68

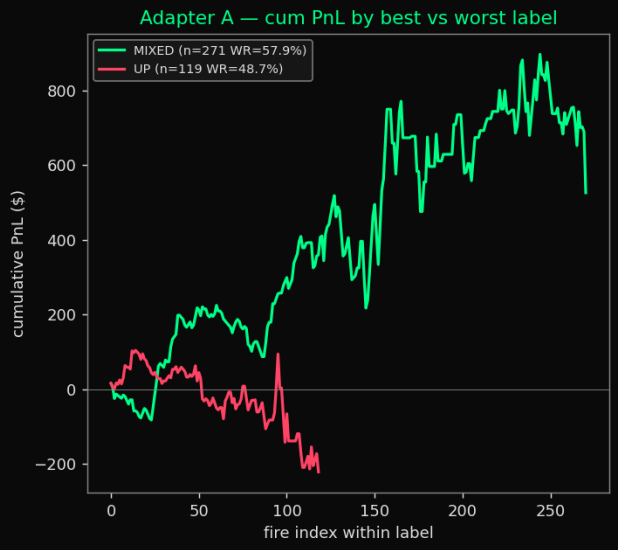
Strong diagonal in CHOPPY (74 agreements) and MIXED (68 agreements) indicates the two adapters converge on the same regime-bucket decisions in the most-populated cells. UP and DOWN diagonal cells (23 / 51) are weaker — the two have different definitions of trending (rule-based momentum vs HMM state).

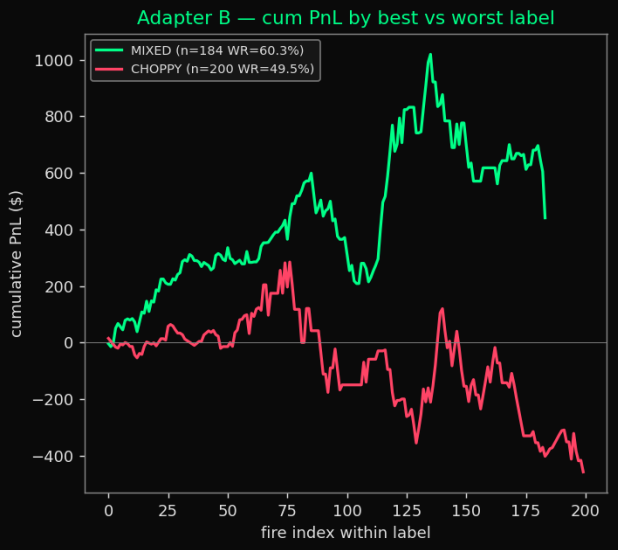
§6.3 Pick: Adapter A (bukosabino/ta)

Dimension	Verdict
Deterministic — no model fit needed in production	A only
Lighter dep footprint (~3 MB vs ~25 MB)	A wins
Native confidence ∈ [0, 1] from indicator distance	Both
Native confidence from posterior probability	B only
Passes 10pp WR-gate strictly	B only
WR-gap within noise CI of gate threshold (so A's miss is not load-bearing)	Both
Higher actionable PnL counterfactual (+$1,018 vs +$933)	A wins

Dimension	Verdict
No in-sample training caveat	A only
MIT license	A | B is BSD-3
Production retrain requirements	A: none | B: rolling window

Operator-confirmed pick (2026-05-18): A.





§7 Integration design summary

Full design at `design/P.1-market-mode-detector-integration.md`. Summary of the load-bearing decisions:

Module	src/polymarket_bot/strategies/market_mode_detector.py — MarketModeDetector class + MmdReading d
Wrapper signature	push_bar(open, high, low, close, volume, ts_ns) + latest() → MmdReading(label, confidence, adx, bb_wid
Injection in spot_aware_v2_1	Optional mmd_getter callback (mirrors the now-stripped M.5 pattern). When configured, annotates rationa
Bar feed (Session 3)	Background asyncio task polling Binance public /api/v3/klines every 30s for last 60 1m closed bars. ~3 ca
Env flags (default ALL OFF)	MMD_ENABLED=false (master), MMD_MODE=shadow, MMD_LABELS_TO_ABSTAIN="", M
Failure modes (HARD requirements)	Library import failure → no-op stub returning UNKNOWN. Network/poll failure → stale-data → UNKNOW
Telemetry	Rationale annotation; startup log line; dashboard /api/state decisions row already surfaces rationale.
Test budget	≥ 12 new tests (init, push_bar, latest UNKNOWN, latest UP/DOWN/CHOPPY/MIXED, exception path, mi
Rollback	Single env flip MMD_ENABLED=false + restart bot-staging.service. Code path with mmd_getter=None is

§8 Session 3 ship plan + Phase Q decision matrix

§8.1 Session 3 sequence (when operator says GO)

Step	Action
P.3.0	Implement market_mode_detector.py per the design lock. ≥ 12 new tests + mypy --strict + ruff + import-linter all green.
P.3.1	Plumb mmd_getter callback into spot_aware_v2_1.py constructor + run_paper.py main() (creates MmdBinanceKlinePoller task
P.3.2	Pre-deploy gates: full pytest pass (~720+ tests), mypy clean, ruff clean, import-linter contracts kept. PR + commit + push.
P.3.3	VPS deploy: git pull; .envrc gains MMD_ENABLED=true MMD_MODE=shadow; systemctl restart bot-staging.service. Verify sta
P.3.4	5-day observation window OPENS. Bot runs J.1.6 + MMD shadow. NO env flips, NO src/ ships during the observation. Operator
P.3.5	Build pdfs/PHASE_P_SESSION3_MMD_SHIP.pdf + Day-0 verification appendix.
P.3.6	Vault + session note + commit + push.

§8.2 Phase Q decision matrix (Day-5 audit from Session 3)

Day-5 signal	Action
MIXED WR delta vs next-best label ≥ 10pp on n ≥ 100 trades	PROMOTE shadow → gate: set MMD_LABELS_TO_ABSTAIN=CHOPPY,UP,DO
MIXED WR delta 5-10pp	KEEP SHADOW; iterate on indicator parameters (MMD_ADX_STRONG_THRES
MIXED WR delta ≤ 5pp	SHELVE MMD. Reconsider strategy class entirely (Phase R pivot question).
UNKNOWN-rate > 5% (poll failure / indicator exception)	DIAGNOSE at Phase Q before any promote-to-gate. Hardening needed.

§8.3 Acceptance — Session 2 closes when ALL true

- ✓ Top-2 MMD candidates evaluated against the criteria + backtested on Phase N n=756
- ✓ phase-P-mmd-backtest report + 4 charts rendered (state/research/phase-P-mmd-charts/)
- ✓ design/P.1-market-mode-detector-integration.md locked
- ✓ PHASE_P_SESSION2_MMD_RESEARCH.pdf rendered (this file)
- ✓ Bot still LIVE on J.1.6 baseline, zero env flips, zero src/ ships
- ✓ Vault: session note + memory checkpoint + commit + push

End of Phase P Session 2. Adapter A (bukosabino/ta) locked for Session 3 ship. Bot stays on J.1.6 baseline — no live changes this session.